Topic 1 End of Topic Assessment: Teacher Notes

This End of Topic Assessment covers **all of Topic 1, Lessons 1–6**, not only Lessons 4–6. If you have taught only Lesson 4, these are the questions your students can do:

- **Lesson 4 question:** Q8.
- **Can be solved with Lesson 4 skills (slope formula from a table + writing an equation):** Q3, Q5, Q9.
- **Partly (L4 skill or prior knowledge, but the context interpretation is L6):** Q11, Q12, Q13.
- **Not yet taught:** Q1, Q6 (L1 regression), Q2, Q4 (L2 correlation), Q7, Q10 (L5), Q14 (L3/L6 domain).

Options: give Q3, Q5, Q8, Q9 (plus Q11–13) now as a Lesson 4 check, and save the rest for after Lessons 5–6 and the Lessons 1–3 review.

Lesson alignment

Q	Skill	Lesson	Can do after L4?	Note
1	Line of best fit (linear regression) from a table	L1	**No**	Least squares regression.
2	Correlation vs. causation	L2	**No**	
3	Rate of change from a table, with units	L3 (also L6)	**Yes**	Same as the slope formula from a table (Activity 4.1).
4	Interpret the correlation coefficient	L2	**No**	$r \approx 0.04$.
5	Write a linear function from a table	L3 (also L6)	**Yes**	Slope from the table plus a known point (L4); the y-intercept is given at $x = 0$.
6	Predict with a regression line	L1	**No**	$y \approx 14.36x + 2.25$.

7	Slope from standard form	L5	**No**	Standard form is introduced in 5.3.
8	Slope + point → equation (point-slope → slope-intercept)	**L4**	**Yes**	The core Lesson 4 skill.
9	Rate of change from a table and interpret it	L3/L4 (interpretation L6)	**Yes**	Slope formula from a table; same context as L4 Getting Started (Draining the Pool).
10	Interpret the x-intercept from a graph	L5 / L6	**No**	"x-intercept" is a new key term in L5.
11	Interpret the y-intercept from a graph	L3 / L6	**Partly**	Students know b from slope-intercept form; interpreting it in context is L6.
12	Slope from a graph and its meaning	L4 slope formula / L6	**Partly**	They can compute it with L4 skills; interpreting it in context is L6.
13	Evaluate a function	Prior (Module 1)	**Partly**	Prior knowledge, not tied to a lesson.
14	Domain of a discrete function	L3 / L6	**No**	Discrete domain ties to sequences (L3).

Answer key (Form M = modified; original letters shown for the standard form)

Q	Form M answer	Original	What was modified
1	B. y = 1750x + 40,500	C	4→3 choices; added an x row (years since 2000); LinReg tip.
2	C. don't brush → cavities	J	4→3 choices (removed beach); causation defined.
3	2.75 dollars per cookie ($2.75 profit for each cookie sold)	—	3-step frame; units supplied.
4	B. Weak positive correlation (r ≈ 0.04)	C	4→3 choices; added a scatter plot; r reminder.
5	y = −77x + 950	—	Added an x row; b/m/equation frame.
6	B. $131.46	H	4→3 choices (removed $34.61); 2-step tip.
7	m = −5 (3y = −15x + 45 → y = −5x + 15)	—	Solve-for-y steps.
8	y = (1/3)x − 5/3	—	Point-slope formula given with the values named; 3-step frame.
9	−60 liters per minute; the water level goes down 60 liters every minute	—	Two points chosen for the student; sentence frame.
10	(40, 0): after 40 minutes Gracie is 0 miles from home, so she is home.	—	Larger graph; grid scale stated; sentence frame.
11	(0, 8/3) ≈ (0, 2.67): she starts 2 2/3 miles from home.	—	Sentence frame; the 1/6-mile grid scale is stated.
12	m = −1/15 mile per minute: she gets 1/15 mile closer to home each minute (4 mph).	—	Two points marked on the graph; sentence frame with a circle choice.
13	B. −94.94	B	4→3 choices (removed −29.9); substitution shown.

14	B. {0, 1, 2, …, 10}	G	4→3 choices (removed 0 ≤ y ≤ 20); domain = x-values reminder.

Accommodations built into Form M (504 / IEP)

- Formula & reminder sheet attached (slope, all three equation forms, intercepts, r, LinReg keystrokes, domain/range, causation).
- Multiple choice reduced from 4 to 3 choices (one distractor removed).
- 14-pt sans-serif font, 1.5 line spacing, one boxed question at a time, numbered steps.
- Key words in bold; vocabulary defined in the question (causation, domain, intercepts).
- Open-response items chunked into numbered steps with fill-in blanks and sentence frames.
- Visual supports: scatter plot added to Q4; Q10–12 graph enlarged, with the grid scale stated and two points marked.
- Added x-rows to tables (Q1, Q5) so students don't have to convert years or months themselves.
- Calculator allowed; work space provided.
- Same 14 items and numbering as the original, so you can grade both forms with one gradebook.

Apply each student's plan alongside this form: extended time, small-group or separate setting, read-aloud of questions (not of numbers in computation items, if your state rules require it), and breaks. Form M keeps the same skills as the original, so it is an accommodated form. If a student's IEP calls for a **modified** assessment with fewer items, give Q3, Q5, Q7, Q8, Q9, Q12 and Q13 (core linear-equation skills).

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